B K TEXTILE installation and backend guide

Prepared for B.K. TEXTILE management and the developer maintaining the software. Handover date: 2 November 2026. Current source release: 2.4.1.

This guide explains how to install, run, maintain and inspect the garment management software. The archive includes the functioning backend code and frontend source. JSON examples are HTTP request bodies, not commands Windows executes by itself. Examples contain fictional values and must not be posted unchanged to business records.

Current deployment and data location

The application runs a Node.js HTTP server and a SQLite database on the host computer. It is not a web-hosted or cloud-synchronized database. By default, the server listens at http://127.0.0.1:4387 and stores records in data/bk-textile.sqlite beside server.mjs. BK_DATA_DIR overrides the data folder; BK_PORT overrides the port. A browser interface does not mean records are on the public web.

Orders, users, hashed passwords, sessions, audit entries, receipts, inspections, bills, payments and report snapshots are in SQLite. Garment photographs are in data/uploads. Original scanned challans are in data/challans; metadata and extracted text are in SQLite. Logs and server.pid are also under data. SQLite WAL/SHM companion files can exist while the app runs. The handover excludes this live data.

The earlier browser preview at 127.0.0.1:5174 and static CB2210 HTML file are historical preview surfaces. The documented current application uses port 4387. Verify /api/version rather than assuming a preview is the installed shared-records software. This handover does not certify which version is currently running on your host.

Staff access inside the office

Start B.K. Textile.cmd launches the normal local-only server. It binds to 127.0.0.1, so other computers cannot reach it. Each employee's localhost refers to that employee's computer, not the central host.

To share records on the office network, first stop the local-only copy, then run Start for office network.cmd on the designated host. It passes --lan and binds to 0.0.0.0. Staff browse to http://HOST-LAN-IP:4387 and use individual department accounts. Possible staff addresses are written to data/server.log. The host must remain awake, running and connected to the same trusted network. Windows Firewall and network policy must permit access. A stable host IP or internal hostname should be arranged by your IT team.

All employees connect to one server and one database. Do not place the SQLite file in a shared folder for employees to open directly. Do not start separate installations expecting automatic synchronization. Stop the old local-only process before changing modes; otherwise the launcher can simply reopen the existing local-only app.

Access outside the office

Outside-office remote access has not been deployed. IT can design private remote-network access to the office server, or move the application to a managed server with HTTPS, access controls, backups and monitoring. These require separate setup; this ZIP does not activate them.

The current application uses plain HTTP and is intended for a trusted private network. Do not expose port 4387 directly to the internet. No public domain, hosted database, automatic cloud backup, external identity service or cloud synchronization was provisioned. No external service is required for the bundled app or local OCR to run.

Archive structure

01 Current software v2.4.1 contains a clean runnable Windows copy, its Node runtime and offline scanner assets. It contains no live database or staff accounts. It will create a fresh data folder when started. Use this for a new installation, not as an accidental replacement for existing records.

02 Historical releases and patches preserves all 11 available release/update ZIPs, including superseded packages. For an existing installation, use BK-TEXTILE-Complete-Update-v2.4.1.zip. Do not apply all old patches sequentially. The original v2.4 small update is retained as history, including its omission of scan assets needed by the older installation.

03 Developer source contains editable backend and React code, build entry, dependency lockfile and tests. It excludes node_modules and working data. 04 Technical files and JSON examples contains requests, commands, route inventory, actual SQLite schema and hashes. 05 Documents contains the Word and Markdown guides and PDF reading copies if rendered. 06 Reference previews contains illustrative screenshots and the initial static programme.

Available history covers initial v1.0, v1.1 and its small update, v1.2 reports, v1.2.1 spinning mills, v2.0 complete material features, v2.1 numbering, v2.1.1 schedule fix, v2.3 department/dashboard features, v2.4 multiple fabrics and v2.4.1 complete updater repair. There is no separately archived v2.2 ZIP; that dashboard work was integrated into v2.3. Exact filenames and SHA256 hashes are included.

Updating an existing installation

Back up the full data folder first. Extract the complete v2.4.1 updater ZIP, keep the existing software running so the updater can identify it, and run UPDATE MY SOFTWARE.cmd. The updater verifies the runtime path, backs up app files, stops the matching process, copies data to its backup, replaces app assets, restarts and checks version/feature flags. Successful completion must report 2.4.1. Refresh the browser with Ctrl+Shift+R.

If asked for the installation folder, select the folder containing Start B.K. Textile.cmd, runtime, public and data. If Windows blocks process stopping, use Stop B.K. Textile.cmd and retry. Do not delete data or select a blank installation as a workaround. Process inspection/stopping can be restricted by Windows account or system policy; the historical stop script uses Windows process inspection.

The corrected complete updater includes OCR assets, uses netstat plus the executable path to identify the running app, and preserves failure diagnostics. Earlier updater failures were reported on the v1.1 installation. Complete payload checks passed in isolation, but successful application to the live host was not confirmed in the last troubleshooting exchange. This archive does not perform an update itself.

New installations and verification

For a new installation, extract folder 01 to a permanent local folder and run Start B.K. Textile.cmd. Create the first administrator, then named staff accounts. No shared default password is supplied. A new folder creates an independent database; moving an existing operation requires deliberate transfer of its complete backed-up data.

Read GET /api/version and check the header. The expected source release is 2.4.1 with the recorded feature flags. verified-version.json was captured from a disposable instance for this handover; it is not a live health report for your host. For LAN sharing, also test from a second staff computer using the host IP.

Backup and recovery

Admin can download the database through GET /api/backup, which checkpoints SQLite first. That download excludes photos and scanned challans. A full business backup needs the entire data folder. For file-level copying, stop the application first so database and companion files are consistent, then copy the complete folder to a protected destination separate from the working computer.

To restore, stop the server, retain the current data folder, restore a chosen complete backup to the same data location, and start a compatible software version. Verify orders, accounts, receipts, inspections, photos, original challans and saved reports before staff resume work. Missing attachments cannot be recreated from the database alone. No automatic backup schedule or disaster recovery service was deployed.

On update failure, rollback attempts to restore app files; it does not automatically replace the working database with the backup. A started newer version may already have created tables. Preserve update-backups and logs for diagnosis. The corrected updater retains failed-start logs and update-error.txt for failures within its guarded update stage.

Backend source map

server.mjs owns HTTP routes, authentication, tables, shared writes, uploads, report snapshots and startup. domain.mjs owns roles, commercial validation, fabric calculations, multiple fabrics, approvals, schedules and invoice helpers. fabric-planning.mjs calculates finished/greige/yarn needs and validates recipes. mill-order-forms.mjs generates spinning, knitting and dyeing order packs.

material-operations.mjs validates receipts and cutting balances. fabric-quality.mjs evaluates custom defects, shrinkage/growth and fastness. mill-finance.mjs evaluates payment terms, eligibility and balances. challan-reader.mjs parses labelled OCR text. delay-tracking.mjs evaluates overdue tasks. department-programmes.mjs owns trims/embellishment/wash catalogues and supplier order forms. production-programmes.mjs generates department programme snapshots.

The React entry is work/desktop-app.jsx. MultiFabric.jsx, FabricCalculator.jsx, ProcessPlanning.jsx, MaterialControl.jsx, ChallanUpload.jsx, ScheduleEditor.jsx, DelayDashboard.jsx, DepartmentProgrammes.jsx, ProductionProgrammes.jsx, SpecImport.jsx and SavedMillOrders.jsx implement the main interfaces. work/build-desktop.mjs emits work/bk-software/public/app.js. Complete backend code is provided, not just excerpts. Backend-source-index.json identifies each module and its hash.

Commands to run the backend

From the current application folder, use one of these alternative launch modes. Do not start both on the same port.

```
& '.\runtime\node.exe' '.\server.mjs'
& '.\runtime\node.exe' '.\server.mjs' --lan
Invoke-RestMethod 'http://127.0.0.1:4387/api/version'
```

BK_DATA_DIR and BK_PORT can select an isolated test folder and different port. They must not point testing tools at production records. The runnable build needs no npm installation. Developer rebuilds need a compatible Node release supporting built-in node:sqlite, and dependencies from the included package.json/lockfile.

From 03 Developer source, a developer can run:

```
npm ci
node work/build-desktop.mjs
node work/test-desktop.mjs
node work/test-materials.mjs
node work/test-departments.mjs
node work/test-multi-fabric.mjs
node work/test-spec-import.mjs
```

The tests create disposable data and use test ports. Test passwords in fixture scripts are not production credentials. Backend-commands.ps1 is commented so opening/running that reference file does not start servers or post records automatically.

JSON request rules

POST requests use Content-Type application/json and an Origin header matching the exact request host and port. Most routes require a signed-in session. Login returns the HttpOnly bk_session cookie. Keep the same hostname consistently; localhost and 127.0.0.1 differ for origin purposes. Do not store real passwords or session cookies in source or examples.

Order updates POST to /api/orders/{orderId} with version, action and data. Read the latest numeric version first. HTTP 409 means another user changed the record; reload and review rather than overwrite it. Roles are enforced server-side. Admin does not bypass department-owned edits and approvals. Authenticated shared reading is broad; this is not a per-field confidentiality system.

```
{
  "version": 1,
  "action": "approve",
  "data": {"section": "fabric"}
}
```

This example requires a Fabrication account and the correct order ID/current version. It approves saved fabric specifications, not a received lot or a payment. Merchandising owns commercial approval and Sampling owns styling approval.

```
{
  "version": 1,
  "action": "programmeInstructions",
  "data": {
    "instructions": {
      "cutting": "Approved marker and bundle instructions"
    }
  }
}
```

This example belongs to Cutting. API-examples.json contains the full structures for orders, fabrics, calculators, measurements, approvals, instructions, schedules, directories, receipts, reconciliation, delay follow-up and printable copies. Replace placeholders and example values. DATA entries are nested objects, not endpoints. No external integration or API-key service is provisioned.

API route inventory

Public/session routes include /api/version, /api/session, /api/setup and /api/login. Shared access includes /api/data, /api/logout, /api/change-password, /api/users and /api/user-access. Order/photo/calendar routes include /api/orders, /api/orders/{id}, /api/photo and /api/calendar.

Fabric/supplier routes include /api/mills, /api/report-copies, /api/department-data, /api/catalog, /api/suppliers, /api/department-documents and /api/production-programmes. Material routes include /api/challan-upload, /api/material-records, /api/operations, /api/inspections, /api/inspection-approve and /api/yarn-receipt-approve. Accounts uses /api/mill-bills, /api/mill-payments and /api/cheque-clear.

Management follow-up uses /api/dashboard, /api/delay-inbox, /api/milestones, /api/milestone-complete, /api/delay-questions, /api/delay-replies and /api/delay-close. /api/audit and /api/backup provide audit and database backup access. ID-specific GET routes return saved reports or attachments. api-route-inventory.json lists paths/methods; server.mjs remains authoritative for fields and role checks, not an autogenerated OpenAPI contract.

Calculations and code behavior

The actual startup selects the listening address as follows. Full source is included alongside the runnable build and under Developer source/work/bk-software.

```
const host = process.argv.includes('--lan')
  ? '0.0.0.0' : '127.0.0.1';
const port = Number(process.env.BK_PORT || 4387);
```

SQLite opens in the chosen data folder with WAL mode. database-schema.sql is captured from a disposable current instance and documents actual tables. Order payloads are JSON inside SQLite rows, not public .json order files.

```
const plannedPieces = Math.ceil(
  buyerQty * (1 + cuttingExtra / 100) - 1e-9
);
const finishedKg = plannedPieces * effectiveKgEach
  * (1 + merchandisingFabricWaste / 100);
const greigeKg = finishedKg / (1 - dyeingLoss / 100);
const yarnKg = greigeKg / (1 - knittingLoss / 100);
```

This simplified explanation is applied per size/fabric and summed. Calculator net kg per piece equals effective width in metres times total fabric length in metres times GSM, divided by 1000 and produced piece count. Tube width is doubled. Calculator waste is applied first; merchandising allowance is additional. Manual kg/dozen is divided by 12. Each fabric line is counted once.

Recipes must total 100 percent. Matching yarn specification, purchase unit, bag weight and spinning mill are pooled before whole-bag rounding. Default bag weight is 45.36 kg; polyester/spandex remain kg. Bag surplus does not increase fabric targets, and existing yarn stock is not automatically deducted.

Invoice allocation runs in a database transaction. The year's series starts above 4055 and the highest saved sequence, giving 4056/BK-initials/year, then 4057, and so on. Customer initials share one sequence. Existing assigned numbers are preserved. Each year's new series starts at 4056 in the current implementation. This is an internal order reference, not a certified statutory tax invoice system.

Operating limits and security

Passwords use salted scrypt hashes. Session tokens are hashed in SQLite and sent as HttpOnly cookies. POST origin, role and stale-version checks are implemented. Plain HTTP does not provide internet transport encryption. Remote production hosting would require a separate deployment and security review.

The scheduler skips Sundays and holidays in the saved reviewed national calendar, refuses uncovered periods, and treats Saturdays as working days. Holidays are not downloaded automatically. The initial reviewed sample window is 27 October 2026 through 15 January 2027, with limited listed dates. Management must review complete coverage, including lunar and newly declared holidays, before releasing plans.

Excel PO extraction suggests recognized fields rather than understanding every buyer layout. Photos are styling references, not reliable sources for composition, GSM, prices or measurements. Measurement import maps worksheets/header rows and long/wide layouts, previews values and checks sizes/duplicates. Styling approval resets when saved specifications change.

Challan OCR runs locally in the browser with bundled PDF/OCR libraries and English/Urdu assets. The UI limits scans to 8 MB and up to six PDF pages. Labelled-field parsing and exact registered mill matches flag ambiguity. Users approve extracted receipts before posting stock. Unfamiliar layouts, handwriting, missing fields and multi-item scans are not guaranteed automatic. Multi-fabric receipts require the exact fabric and distinct lot references.

QC finalization can make Accounts eligible to record payment under saved terms. Issuing or clearing a cheque records bookkeeping state; it does not transfer money, print a bank cheque or connect to a bank. Partial balances, cheque issuance and clearance are separate. The app is operational tracking, not a full audited general ledger or payroll system.

Reports remain drafts until data and approvals are present. Printing does not move stock, send a supplier order, approve styling or authorize payment. Saved copies retain their source revision. Delay follow-up stays in the app and does not send email, SMS or WhatsApp messages.

Verification and next deployment steps

Checks cover department permissions, stale writes, units/wastage, schedule loops/holidays, independent orders, numbering, yarn recipes/bags, supplier print snapshots, receipts/QC/payments, cutting balances, catalogues, multiple fabrics and spec parsing. The complete 2.4.1 payload was started against disposable data for this handover and its feature flags verified. This is not proof of successful installation on the live host.

Before rollout, confirm the live version and data folder, test actual buyer/spec/challan formats, review holiday coverage, test a second LAN device, assign backup ownership and test recovery. Department heads should review report inputs and limits. Decide separately whether outside-office access is needed and how IT will deploy it.